

Should the Urgency of Visual-Tactile Takeover Requests Match the Criticality of Takeover Situations?

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Abstract—Highly-automated driving requires drivers to take over the control of their vehicle in any situation. In takeover situations with small time headways, a fast takeover is crucial. Here, drivers could be supported by an appropriate takeover request. This article presents a driving simulator study investigating whether less critical takeover situations and a match between the urgency of a takeover request and the criticality of the takeover situation benefits drivers regarding their takeover performance and subjective experience. Fifty-two participants experienced two takeover situations with varying time headways (small and large). A low urgent, visual-tactile takeover-request was presented to half of the participants. The other half experienced a highly urgent, visual-tactile takeover request, resulting in two conditions, in which the takeover requests matched the criticality of the takeover situation, and in two conditions, in which the takeover requests did not match. The results showed that participants generally performed better in low critical takeover situations, but they took over quicker in highly critical situations. However, a match neither shortens takeover time nor enhances subjective experience.

Index Terms—Automation, driver behavior, driver performance, driving simulator, human factors, human-computer interaction, intelligent vehicles.

I. INTRODUCTION

IN THE past years, automated driving has gained much attention and its development progressed remarkably. A highly automated vehicle [1], also known as SAE level 3 “Conditional Automation” [2], takes over the vehicle’s longitudinal and lateral control and monitors the environment. In case the automation reaches its system limits, the driver is required to take back full control over the vehicle on request. The characteristics of these takeover situations are diverse, they may differ regarding time headway, predictability, traffic, etc. Unpredictable takeover situations with small time headways, so-called “non-scheduled system initiated” [3] or “unstructured” [4] transitions, will be very rare but are highly critical. The driver switches from being “out-of-the-loop” to the operator [5]. Situation awareness has to be developed as fast as possible, hence, the drivers’ visual

and cognitive demand increases quickly. This task switch takes time while fast takeovers are required. In this situation, a clear driver-vehicle-interaction is even more important. An appropriate takeover request-design, for example a more or less urgent design depending on the situation, may support the drivers’ cognitive processing and enhance his or her takeover performance [6]. We, therefore, focus on matching the urgency of takeover requests (TOR) to the criticality of the takeover situation for non-scheduled system initiated transitions.

A. Criticality of the Takeover Situation

In takeover situations, the driver has to perceive and comprehend the driving situation and to select and execute an action within seconds [6]. A decrease in time headway leads to an increase of criticality of the takeover situation. Damböck *et al.* [7] compared the takeover quality in situations with three different time headways and complexities. In lane change situations, drivers’ error rate was lower and equal to manual driving with a time headway of 8 s than with smaller time headways of 6 s or 4 s [7]. In a similar driving simulator study, drivers deviated less from the lane, showed better steering behavior and fewer collisions with time headways of 8 s and 5 s than with 2 s [4]. Regarding takeover time, brake and steer input times were found to be longer with 7 s than with 5 s time headway, even though gaze reaction time did not differ [8]. In contrast, Walch *et al.* [9] did not find a difference in driver behavior between 4 s and 6 s time headways in their driving simulator study. To conclude, larger time headways improve takeover quality but delay drivers’ responses to TORs. And in takeover situations with time headways of 8 s or more, drivers’ takeover quality is comparable to manual driving.

B. Takeover Request Design

Amongst others, the design of a TOR is determined by the modalities that present the request and by the conveyed urgency of the request.

1) *Modality*: Takeover requests can be presented via auditory, tactile or visual cues or in combination. Auditory and visual cues may be abstract (e.g., single tones) or language-based (e.g., speech). One benefit of auditory and tactile information is that they can be perceived irrespective of the visual focus of the driver. Another benefit is that their perception requires different perceptual modalities than the perception of the driving

Manuscript received August 10, 2018; revised February 4, 2019 and July 29, 2019; accepted November 18, 2019. Date of publication November 25, 2019; date of current version May 25, 2020. (Corresponding author: Fabienne Roche.)

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Digital Object Identifier 10.1109/TIV.2019.2955906

scene. Based on Wicken's Multiple Resource Theory, this should result in better performance because these cues can be processed without impairing the driving performance [10]. However, visual information, presented in the windscreen and related to the driving scene (e.g., contact analog head-up display or augmented reality), might support the drivers to develop situation awareness [11]. Damböck *et al.* [11] compared the drivers' failure detection and takeover performance with and without a contact analog head-up display. When supported with information via head-up display, participants detected system failures earlier and deviated less from the lane than without the additional information [11]. In a series of studies, Politis *et al.* [12]–[16] compared drivers' reaction times and perceived urgency of auditory, tactile and visual TORs and all possible combinations. An abstract visual TOR (red circle) presented on the driving scene (comparable to a head-up display) resulted in faster takeovers than an abstract auditory or tactile TOR [12]. Various studies consistently found faster reactions with bi- and trimodal stimuli compared to unimodal stimuli, e.g., faster reactions with a visual-tactile than a visual TOR [13], [16]–[19]. Furthermore, those bi- and trimodal TORs are perceived as more urgent than unimodal TORs [12], [14], [15]. Another study by Politis *et al.* [16] showed that abstract cues and visual cues are recognized faster than language-based and auditory takeover requests. Hence, a combination of abstract visual cues with tactile or auditory cues promises fast takeovers and high perceived urgency. Thereby, the visual cue should be presented in the windscreen. This, in turn, might lead to more appropriate and faster driver responses to TORs in critical takeover situations.

2) *Urgency*: Prior work showed that pulse rate [12], [13], pulse duration [20], color [12], [13], [20]–[22], and wording of a speech message [14], [15] influence the perceived urgency of a TOR and the drivers' takeover performance. For abstract visual cues, higher pulse rates and the colors red and yellow were perceived more urgent [12], [21] and resulted in faster takeovers [13] than lower pulse rates and the color green. Lorenz *et al.* [22] investigated if the drivers' takeover behavior can be quantitatively and qualitatively influenced by the information presented via a head-up display. In a "non-scheduled" takeover situation, a red overlay over the drivers' blocked lane resulted in more brake-only reactions and descriptively stronger brake reactions than a green overlay over the free trajectory or a generic takeover request without overlay [22]. The takeover times did not differ [22]. The color red seemed to convey higher urgency and led to braking rather than steering, but could be prone to very strong driver reactions. Borojeni *et al.* [20] varied TOR-urgency and road curvature. They found that participants take over faster with an urgent auditory-visual TOR on straight roads. To sum up, the modality and the urgency of a TOR influence the drivers' cognitive processing of the driving scene and hence the takeover behavior.

C. Objectives

Since fast and safe takeovers are crucial in time-critical takeover situations and since research showed that urgent TORs/warnings may result in faster takeovers, the present paper

investigates the effects of visual-tactile TOR-designs, whose urgencies did or did not match the criticality of takeover situations. We designed two visual-tactile TORs, one communicating low urgency (low urgent TOR-design), the other one communicating high urgency (highly urgent TOR-design). We presented them in takeover situations with either low or high criticality (time headway). This resulted in two conditions, in which the TOR-design's urgency matched the criticality of the takeover situation, e.g., low urgent TOR-design + low critical takeover situation, and in two conditions, in which the TOR-design's urgency did not match the criticality of the takeover situation, e.g., low urgent TOR-design + highly critical takeover situation. Therefore, the study aims at demonstrating the advantage of large time headways and at assessing possible benefits of situation-dependent (i.e., adaptive) TORs. The idea is that takeover situations with small time headways could be compensated by faster takeovers, which are facilitated through highly urgent TORs. And vice versa, in low critical situations, a low urgent TOR could arouse better performance. To date, no study about situation-adaptive TORs has been published yet. However, results from aviation research indicate that decision systems, allocating the authority between system and pilot based on the situation, result in better performance than non-adaptive systems [23].

D. Hypotheses

First, we expected longer takeover times, smaller steering wheel positions, smaller brake pedal positions, fewer brake reactions, fewer collisions, and lower ratings of perceived effort in the low critical compared to the highly critical takeover situation. Second, for the low urgent compared to the highly urgent TOR-design, we expected smaller steering wheel positions, smaller brake pedal positions, and fewer brake reactions. Third, we hypothesized that the participants' perceived effort is lower and their acceptance of the highly-automated system is higher with a TOR-design that matches the criticality of the takeover situation than with a TOR-design that does not match. Forth, we expected shorter takeover times and more brake reactions in the highly critical situation with the highly urgent TOR (highly critical match) compared to all other matches, and vice versa in the low critical situation with the low urgent TOR (low critical match).

II. METHOD

A. Participants

Fifty-two participants (24 women) aged between 21 and 63 years ($M = 27.2$ years, $SD = 6.0$) took part in this study. All participants held a valid driver's license between 1 and 45 years and reported to use the car between never and six times per week ($M = 1.1$, $SD = 1.4$). Some of them received course credit for their participation. The study was approved by the local ethics committee.

B. Materials

The experiment was conducted in a mid-fidelity *driving simulator* consisting of a Volkswagen vehicle mock-up including

Fanatec¹ pedals, a Fanatec steering wheel, cockpit, seat, gear shift, and in-car entertainment systems. OpenDS² 4.5 was utilized to simulate a two-lane rural road including a crash barrier and other vehicles on both lanes. The driving scene was projected on a screen 80 cm in front of the vehicle mock-up. The size of the projection screen was 3 m × 1.70 m, enabling a field of view of about 124°. A rear-view mirror was embedded in the projection.

The *criticality rating* of the takeover situation was assessed using a unidimensional 8-point scale, ranging from 0 (not critical) to 7 (very critical). This scale was developed in-house and used in previous studies [24]. It was used to check whether the manipulation of the criticality of the takeover situation was perceived by the participants.

The unidimensional scale of *perceived effort* by [25] was used to assess participants' perceived effort. It shows a 10 cm vertical line with 12 ticks on the left and 7 ticks on the right side. Ticks on the left side were evenly distributed and labeled with ascending numbers ranging from 0 to 220 (max. effort). Ticks on the right side were labeled using a 7-point Likert scale ranging from "hardly effortful" to "extraordinary effortful". Participants were asked to indicate how effortful they experienced the takeover situation by making a cross at the corresponding point of the vertical line. The scale of perceived effort has proven good reliability and validity [25].

An *acceptance* scale measured the perceived usefulness and satisfaction of new information technology [26]. It consists of nine items using five-point rating-scales from -2 to +2. Six items were mirrored. Each item was a semantic differential, five items loading on usefulness and four on satisfaction. Higher scores represent higher perceived usefulness or satisfaction.

Participants were asked to perform a *non-driving related task* (NDRT), which was a questions-and-answers-game. It consisted of questions with three response options each, presented on a tablet PC (Samsung Galaxy Note 10.1, standard factory settings). Participants selected their answers by tapping on it. The next question was presented after choosing one of the answers (or a maximum of 20 seconds). This task requested cognitive, manual and visual resources similar to interacting with a smartphone.

The *visual-tactile takeover request* was a visual cue accompanied by a vibration in the participants' seat. The visual cue was an overlay on the participants' lane mimicking a contact analog head-up display, similar to [11], [22]. We used abstract visual cues because they can be identified quicker than language-based visual cues [12], [16]. The two levels of urgency were realized by varying the color and the position of the visual overlay. The low urgent TOR was realized by a green overlay projected on the free lane next to the participants' lane. The highly urgent TOR utilized a red overlay, which was projected on the restricted participants' lane, similar to Lorenz *et al.* [22], see Fig. 1. The vibrations were presented through an Arduino-Microcontroller at four positions on the participants' seat for 2 s with a frequency of 200 Hz together with the onset of the visual cue. In line with Politis *et al.* [13], the frequency of the vibrations did not differ between the



Fig. 1. Visualization of the two visual TOR-designs with high (red overlay on participants' lane, left) and low urgency (green overlay over the free lane, right).

TOR-designs. The vibration was added because visual-tactile TORs are perceived as more urgent than visual or tactile TORs [12], [14], [15] and produce faster responses than visual TORs [13], [16]. Prior to the main experiment, we surveyed the perceived urgency of the two TOR-designs in a pre-test with 12 participants. There was a significant difference between the low urgent and the highly urgent TOR-design, $t(8) = 3.21$, $p < 0.05$, *Cohen's d* = 1.97.

C. Experimental Design and Procedure

Participants were welcomed at the laboratory and introduced to the experiment. They were asked to provide their demographical data and to sign the informed consent. In the familiarization phase, participants engaged in the NDRT for 10 min, then they manually drove a 2 min practice track in the driving simulator.

In the test phase, participants were informed that the driving automation takes over longitudinal and lateral control of the vehicle once it is switched on (i.e., SAE-level 3 [2]). They were instructed to behave as they would in the real world and that they may be requested to take back control in case the automated system reaches its limits, for example, due to a hazard on its lane. All participants completed two studies in a row, one examining the effects of different visual-tactile TORs on driving behavior and one testing the effects of auditory-tactile TORs. The order of studies was counterbalanced across participants. Because the visual-tactile TORs conveyed different information than the auditory-tactile TORs, the results of the visual-tactile TORs are presented, only. The results of the auditory-tactile TORs are presented in another publication [24]. Participants drove three trials for the visual-tactile TORs.

Participants experienced either the green (low urgent) ($n_1 = 28$) or the red (highly urgent) ($n_2 = 24$) TOR, once in a low critical, once in a highly critical takeover situation. The criticality of the takeover situation was varied by the available time headway when presenting the TOR before reaching the system limits: 7 s for the low critical and 3 s for the highly critical situation. Those values were identified to be the maximum and minimum time headways most frequently used to create non-scheduled system initiated transitions of different criticality [22], [27]–[29]. The design was, therefore, a 2 (TOR-design) × 2 (time headway) mixed design. This resulted in four conditions, two in which the TOR-designs' urgencies matched the criticality of the takeover situation, and two in which the TOR-designs' urgencies did not match (see Table I). We presented the TOR-designs between-subjects because, in real driving situations, drivers will experience one variant without having a reference. In addition, we intended to avoid adaptation effects over the repeated experience of takeover situations, as shown in [30]. Time headway was a within-subjects factor. Additionally, participants experienced a filler trial without a takeover situation (no takeover request, no

¹<https://www.fanatec.com/eu-en/>

²<https://www.opensds.eu/>

TABLE I
EXPERIMENTAL DESIGN: 2 (TOR-DESIGN) $\times$ 2 (TIME HEADWAY)

		Time Headway	
		7 s	3 s
		(low critical)	(highly critical)
TOR-Design	Green (low urgent)	a) Low critical match	b) No match
	Red (highly urgent)	c) No match	d) Highly critical match
$n_1 = 28$			
$n_2 = 24$			

critical situation). The order of trials was counterbalanced across participants.

Each trial started with the automatic acceleration of the simulated car to 100 km/h on the right lane. The test track was a straight, two-lane road with medium traffic passing the participants' car on the left lane. Upon request by the experimenter, participants started to engage in the NDRT. In the experimental trials, after about 1:30 min, the takeover request was issued, because of a broken tractor on the participants' lane. No other traffic was present when the TOR was issued. The participants had to take over control and steer to the left or brake to avoid a collision. In the filler trials, participants drove highly automated for about 1:40 min. In the experimental and the filler trials, they were then asked to stop the car and to rate the criticality of the takeover situation, their perceived effort [25], and the acceptance [26]. The experiment lasted for about 90 min. After each trial, participants had the option to take a break.

Dependent variables included takeover time after TOR (TOT) in seconds, reaction type, number of collisions, the maximum steering wheel position (range $\pm 100\%$, positive values represent steering to the left), maximum brake pedal position (range 0% to 100%), criticality ratings [0–7], perceived effort ratings [0–220], and acceptance ratings [–2 to +2]. A takeover was specified as the first input by the participant regarding a steering wheel position larger than $\pm 4\%$, or a brake pedal position larger than 0.1%, similar to [28]. As soon as one of these thresholds was exceeded, input was considered to be deliberate and takeover time was measured. Takeovers before the TOR-presentation were excluded from the analysis. The variable “reaction type” describes if the participant steered only, braked only or steered and braked between the TOR and the hazard. Maximum brake pedal position and maximum steering wheel position were computed for the period between the participant' takeover and the hazard. Shorter TOTs, fewer collisions, smaller steering wheel positions, and smaller brake pedal positions are defined as better takeover performance.

III. RESULTS

A series of 2 (TOR-design) $\times$ 2 (time headway) ANOVA was computed for each continuous dependent variable with R version 3.4.1 [31] using the afex-package function aov_ez [32]. Effect sizes are reported using partial η^2 [33] with $\eta_{\text{part}}^2 < 0.06$ indicating small effects, $0.06 < \eta_{\text{part}}^2 < 0.14$ medium effects, and $\eta_{\text{part}}^2 > 0.14$ large effects [34]. For the binary variables reaction type and collision rate, general mixed effect models

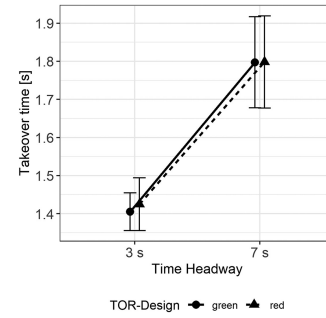


Fig. 2. Mean takeover times for takeover situations with 3 s and 7 s time headways with the green (low) or the red (highly urgent) TOR-design. Error bars ± 1 SE.

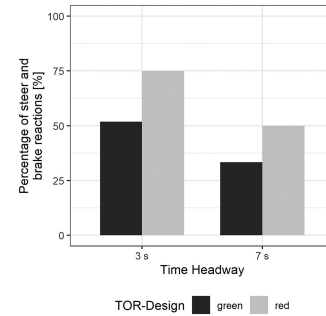


Fig. 3. Percentage of trials in which participants steered and braked in takeover situations with 3 and 7 s time headways with the green (low) or the red (highly urgent) TOR-design. The remaining percentage of the data steered only.

were fitted with the “lme4”- package [35]. The goodness-of-fit of these models was based on two coefficients: the marginal and conditional coefficient of determination (R^2) [36]. Descriptive statistics are presented in Table II, statistical results in Tables III and IV.

In total, 104 trials were recorded, with fifty-two participants experiencing two test trials each. One participant with the green TOR took over control before a TOR was issued in both trials and was therefore excluded from the analysis. This resulted in 102 trials with valid driving data; 52 the with the green TOR, 48 with the red one.

A. Takeover Behavior and Performance

1) *Takeover Time*: There was a highly significant main effect of the time headway on takeover time but no interaction. In the 3 s situation, takeover times were shorter than in the 7 s situation, see Fig. 2 and Tables II and III.

2) *Reaction Type*: In 49 trials (48%), participants steered-only upon takeover request and in 53 trials (52%), they steered and braked. In no trial, participants braked-only. The results of the general mixed effects model revealed that the effect of time headway on reaction type missed significance level (see Fig. 3 and Table IV for detailed results).

3) *Number of Collisions*: The analysis revealed that in five trials (4.9%), participants collided with the tractor or the barrier, all in the 3 s situation. Three collisions occurred with the red TOR, two with the green one. These were too few occurrences to fit a model.

TABLE II
MEANS AND STANDARD DEVIATIONS OF ALL DEPENDENT VARIABLES FOR ALL EXPERIMENTAL CONDITIONS

Measure	TOR-Design M (SD)			
	Green (low urgent)		Red (highly urgent)	
	7 s	3 s	7 s	3 s
Takeover time [s]	1.80 (0.62)	1.41 (0.26)	1.80 (0.59)	1.42 (0.34)
Reaction type [% steered and braked]	33.3	51.9	50.0	75.0
Collision frequency [%]	0	7.4	0	12.5
Steering wheel position [± 100 %]	15.5 (7.9)	65.4 (23.8)	19.7 (11.5)	68.6 (26.1)
Brake pedal position [%]	8.61 (14.0)	25.7 (32.3)	26.4 (33.6)	49.8 (41.9)
Criticality rating [0-7]	3.46 (1.90)	6.25 (1.21)	3.79 (1.79)	6.00 (1.10)
Perceived effort rating [0-220]	53.5 (37.4)	84.8 (46.2)	56.0 (41.3)	79.8 (41.3)
Acceptance: usefulness rating [-2 to +2]	0.31 (0.65)	-0.04 (0.74)	0.47 (0.57)	0.26 (0.65)
Acceptance: satisfaction rating [-2 to +2]	0.38 (0.84)	-0.04 (0.83)	0.32 (0.78)	-0.12 (0.75)

TABLE III
STATISTICAL RESULTS OF ANALYSIS OF VARIANCES

Measure	Main effect TOR-Design			Main effect Time Headway			Interaction TOR-Design x Time Headway		
	F	p	η^2	F	p	η^2	F	p	η^2
Takeover time [s]	0.01 ¹	0.918	0.00	18.66 ¹	< 0.001 ***	0.28	0.01 ¹	0.916	0.00
Steering wheel position [± 100 %]	0.78 ¹	0.382	0.02	239.38 ¹	< 0.001 ***	0.83	0.03 ¹	0.875	0.00
Brake pedal position [%]	7.23 ¹	0.010 **	0.13	22.16 ¹	< 0.001 ***	0.31	0.53 ¹	0.469	0.01
Criticality rating [0-7]	0.01 ²	0.915	0.00	116.21 ²	< 0.001 ***	0.70	1.55 ²	0.218	0.03
Perceived effort [0-220]	0.01 ²	0.91	0.00	27.33 ²	< 0.001 ***	0.35	0.51 ²	0.478	0.01
Acceptance: usefulness [-2 to +2]	0.44 ²	0.508	0.01	8.02 ²	0.007 **	0.14	0.52 ²	0.476	0.01
Acceptance: satisfaction [-2 to +2]	1.39 ²	0.244	0.01	18.55 ²	< 0.001 ***	0.14	0.02 ²	0.892	0.01

Note: ¹df (1, 49), ²df (1, 50). Significance symbols: '.' 0.1, '*' $p < 0.05$, '**' $p < 0.01$, '***' $p < 0.001$.

TABLE IV
STATISTICAL RESULTS OF GENERAL MIXED EFFECTS MODEL

General mixed effects model	TOR-Design				Time Headway				Interaction: TOR-Design x Time Headway			
	β	SE	z	p	β	SE	z	p	β	SE	z	p
Reaction type [% steered and braked]	1.90	1.17	1.62	0.105	-1.37	0.81	-1.70	0.089	-0.62	1.16	-0.54	0.593
Variance explained: $R^2_{\text{marginal}} = 13.5$ %, $R^2_{\text{conditional}} = 60.2$ %												

Note: Significance symbols: '.' 0.1, '*' $p < 0.05$, '**' $p < 0.01$, '***' $p < 0.001$.

4) *Maximum Steering Wheel Position:* The analysis revealed a highly significant main effect of time headway on participants' maximum steering wheel position. In the 3 s condition, the maximum steering wheel positions were larger and hence more extreme than in the 7 s situation, see Fig. 4(a) and Tables II and III for detailed results.

5) *Maximum Brake Pedal Position:* The analysis showed a highly significant main effect of time headway on participants' maximum brake pedal position. In the 3 s situation, maximum brake pedal positions were larger, than in the 7 s situation (see Fig. 4(b) and Tables II and III). Consider that 63% of participants in the 3 s situation and 41% in the 7 s situation steered and

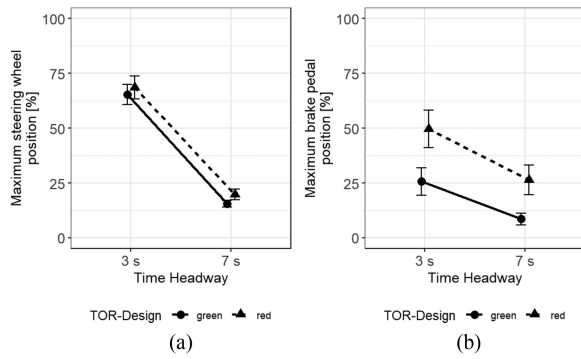


Fig. 4. (a) Mean maximum steering wheel position and (b) mean maximum brake pedal position in takeover situations with 3 s and 7 s time headways with the green (low) or the red (highly urgent) TOR-design. Error bars ± 1 SE.

braked. In addition, a significant main effect of TOR-design was obtained. With the red TOR, maximum brake pedal positions were larger, than with the green TOR. Again, a higher percentage of participants with the red TOR (59%) steered and braked than with the green TOR (45%).

B. Subjective Measures

1) *Criticality Ratings*: There was a highly significant main effect of time headway on criticality ratings. The 3 s situation was rated significantly more critical than the 7 s situation, see Fig. 5(a) and Tables II and III.

2) *Perceived Effort Ratings*: There was a highly significant main effect of time headway on participants' perceived effort ratings. In the 3 s situation, participants' perceived effort was significantly higher than in the 7 s situation, see Fig. 5(b) and Tables II and III.

3) *Acceptance Ratings*: The acceptance ratings were divided into usefulness and satisfaction scores. There was a highly significant main effect of time headway on usefulness and satisfaction scores. In the 3 s condition, usefulness and satisfaction ratings were smaller, than in the 7 s situation, see Tables II and III, Fig. 6(a) and 6(b).

IV. DISCUSSION

This study investigated the effects of the criticality of the takeover situation on takeover quality and of TOR-urgency, whose levels did or did not match the criticality of the respective takeover situation. Criticality of the takeover situation and TOR-urgency were varied on two levels each and all possible combinations were tested in a driving simulator. This resulted in two conditions, in which the urgency matched the criticality of the takeover situation, and in two conditions, in which the urgency did not match. Participants were assigned to either the low urgent (green) or the highly urgent (red) TOR-design and experienced one low critical (7 s time headway) and one highly critical (3 s time headway) takeover situation. Driving data and questionnaires at the end of each trial were collected and analyzed.

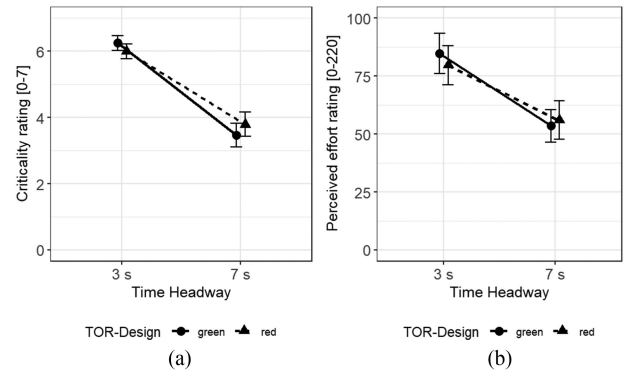


Fig. 5. (a) Mean criticality ratings and (b) mean perceived effort ratings after experiencing takeover situations with 3 s and 7 s time headways with the green (low) or the red (highly urgent) TOR-design. Error bars ± 1 SE.

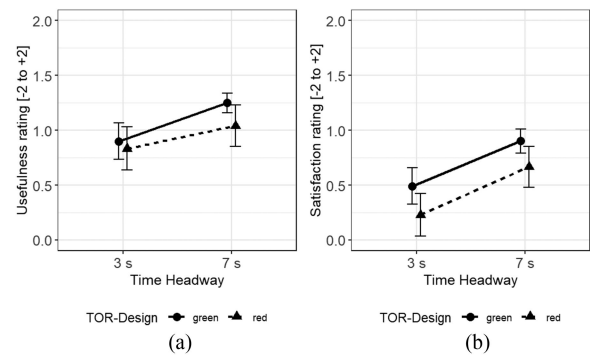


Fig. 6. Mean acceptance ratings divided into (a) usefulness ratings and (b) satisfaction ratings after takeover situations with 3 s and 7 s time headways with the green (low) or the red (highly urgent) TOR-design. Error bars ± 1 SE.

A. Effects of Criticality of the Takeover Situation

Higher criticality ratings for the 3 s situation confirmed that those situations were perceived as more critical than the 7 s situation. The results of all dependent variables supported our first set of hypotheses that in low critical takeover situations, participants take over slower, but show safer takeover performance and lower ratings of perceived effort. This is partially in line with Mok *et al.* [4], who found better steering behavior with 5 s and 8 s than with 2 s time headway. This finding demonstrates that the time headway between TOR and system limits is crucial regarding safety-relevant behavior. However, as expected, takeover times were shorter in the highly critical situation (3 s) than in the low critical situation (7 s), contrasting Walch *et al.* [9], who did not find a difference between small and large time headways. This may be due to the magnitude of the time headways, which were used in the present study. Here, the difference between small and large time headways was larger than in the study of Walch *et al.* [9] (4 s and 6 s). Still, Gold *et al.* [8] found shorter brake and steer input times with 5 s than with 7 s time headway, even though their time headway difference was the same as in Walch *et al.* [9]. As expected, perceived effort ratings were higher in highly critical situations than in low critical situations. Similarly, Damböck *et al.* [7] found lower comfort ratings in the highly critical situation of 4 s compared to 6 s and 8 s. The

hypothesis regarding collisions could not be tested because of a small number of collisions. The frequency of brake reactions did not differ between the two time headways. The results of the present study emphasize the importance of an early warning of the driver. However, they are promising because smaller time headways demand faster takeovers to handle the situation, a behavior that we found.

Surprisingly, acceptance ratings were lower in the highly critical situation, even though they survey the participants' opinion on new information technology itself, and do not intend to measure a judgment of a situation. However, no rating differences were found between the two TOR-designs, which were the intended subject of evaluation. This might be due to the participants' erroneous attribution of the experienced effort in the highly critical situation to the judgment of the system.

B. Effects of Takeover Request-Urgency

We found only one main effect of the TOR-design supporting our second set of hypotheses. Participants with the highly urgent TOR (red) braked more extreme than those with the low urgent TOR (green). This is partially in line with Lorenz *et al.* [22] who used similar TOR-designs and found higher decelerations with the highly urgent TOR-design on a descriptive level. The small effects of TOR-design might be due to too small differences of the urgency between the two designs. TOR-designs that differ more in respect to their urgency (e.g., flashing visual cues) might affect participants' performance more.

C. Effects of Matching the Urgency of the Takeover Request to the Criticality of the Takeover Situation

The results did not support our third set of hypotheses assuming that matches of criticality of the takeover situation and TOR-design result in lower perceived effort and higher acceptance ratings than no matches. Similarly, the results did not support our fourth set of hypotheses expecting shorter takeover times and more brake reactions with the highly urgent TOR in a highly critical takeover situation compared to all other combinations and vice versa for the low critical match.

There are different possible explanations for the lack of expected results. First, in the 3 s time headway condition, takeover times were already very short compared to other studies [27]. Possibly the highly urgent TOR-design could not foster even faster reactions (floor effect). Second, it may be because the TOR-designs were presented between-subjects, hence, intrapersonal differences are missing. Third, it is possible that the used designs were not sufficient in communicating different levels of urgency and to elicit faster takeovers. A flashing visual cue like in [20] or vibration with higher frequencies might support the communication of higher urgencies. For example, Borojeni *et al.* [20] found faster takeovers with the highly urgent than with the low urgent TOR. They used auditory-visual TORs with red flashes for high urgency and yellow flashes for low urgency. Another design option might be the use of different modalities for different urgencies [37]. However, the pretest showed a difference regarding the perceived urgency of the TOR-designs but maybe this perceived difference did not affect takeover behavior. Fourth, another reason might be that the TOR-design

does not influence the takeover time but rather the reaction type. In Lorenz *et al.* [22], takeover times did not differ between the red and green TOR, but more participants braked with the red TOR. Indeed, on a descriptive level, we found more brake reactions in addition to steering with the highly urgent TOR compared to the low urgent TOR.

D. Limitations and Future Work

The study has limitations that should be considered. First, it was conducted in a mid-fidelity simulator. It should be taken into account that driving behavior in real traffic might be different from the one shown in the simulator. Hence, the results might differ from the results of on-road studies.

Second, the periods of autonomous driving before a takeover request (1:30 min) were shorter than they might be in everyday traffic. The results of Brandenburg and Skottke [38] indicated that longer periods of highly-automated driving might also lead to prolonged phases of decreased driving performance after resuming the control of their vehicle. Besides, the same timing of takeover requests in all trials may have increased their expectancy. This might limit the validity of the results.

Third, the used TOR-designs conveyed different information: the green one visualized the free lane, the red one the blocked lane. Hence, not only the urgency varied between both designs.

Forth, the present study tested only two situations which varied in respect to time headway. However, as argued in the introduction, every takeover situation differs from the other. Also, 3 s and 7 s time headway are quite small imposing time pressure on participants. The choice of time headways might have influenced driving behavior. Thus, we covered only a small part of possible takeover situations. Further research could investigate a wider variety of takeover situations. In addition, applying eye-tracking could provide further insights into takeover behavior after a TOR was issued. Time headway and TOR-design might affect participants' information search behavior.

E. Application

The results of the present study show that larger time headways strongly improve takeover performance, but extend takeover times. Therefore, effort should be put in presenting takeover requests in system initiated takeovers as early as possible. However, the results of the present study indicate that matching the urgency of a takeover request to the criticality of the takeover situation does not improve the takeover performance – at least with the TOR-designs we used. Future work should investigate the effect of other TOR-designs that convey different urgencies and the same information. The results of both studies on TOR-urgency suggest that factors of the takeover situation might affect takeover behavior more than the design of takeover requests, at least for non-scheduled takeover requests in time-critical situations [24]. Also, not only time headway and TOR-design should be considered when designing TORs in order to maximize safe takeover behavior. These are for example traffic on a neighboring lane [28], road curvature [20], vehicle dynamics, participants' situation awareness, and driving skills. The results of the present study are a starting point for future investigations.

ACKNOWLEDGMENT

The authors would like to thank V. Jeske and S. Leipold for the careful data collection and M. Lasch for his technical support.

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